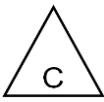


Special characteristics supplier																																													
Symbol	Definition		Supplier																																										
      	When to apply a S? Significant characteristic Product features or process features, which affect the fit, function, performance, following processing	General	SC's (S1-S7) have to be marked continuously in the FMEA's and control plan. (Basically Melecs specified symbol has to be used unless Melecs defined in the technical documents other symbols) SC's (S1-S7) have to be evaluated in FMEA acc. newest handbook rules. Capability and measurement system analyses MSAs of associated measuring equipment must be enclosed to the initial sampling report to Melecs Deviation approval possible in case out of spec.? Yes, temporarily or acc. approval of customer When suppliers have their own special characteristic classification, then a translation table (supplier classification vs. Melecs/customer classification) has to be available																																										
		Initial sampling*	<u>Testing methode SPC:</u> Requirements of process capability/machine capability pending on the SC defined level (x-7) General requirements to capabilities: pp, ppk ≥ 1,67 / cm, cmk ≥ 1,67, min. 50 samples in a row (per cavity) Is the process not capable then it have to be a 100% check until the process has been optimised. Confirmation by CoC (Certificate of Conformity) document with each delivery (after green PPAP). Measurement system analyses/measurement process analyses has to be performed according VDA Vol. 2 or MSA last revision, arranged with responsible SQA. cg, cgk ≥ 1,33 / %R&R <10% (procedure depends on the measuring process)																																										
		Serial production*	<u>Testing methode SPC:</u> Requirements of process capability/machine capability pending on the SC defined level (x-7) General requirements to capabilities: cp, cpk ≥ 1,33, min. 125 pcs., 25 spot check to 5 pcs. (per cavity) <u>Testing methode SPC:</u> In production a SPC control of characteristic has to be performed long-term process capability <u>Poka-Yoke if capable</u> Generally to all testing methodes records have to be made and if necessary send to MELECS, also applies for																																										
		CC/SC Table with requirements:																																											
		<table><tr><td rowspan="11">Significant characteristics</td><td></td><td>100% (in prod. line)</td><td>100% (in prod. line)</td><td></td></tr><tr><td></td><td>Ppk ≥ 2,00</td><td>Cpk ≥ 1,67</td><td></td></tr><tr><td></td><td>Ppk ≥ 1,67</td><td>attributive (gauge)</td><td></td></tr><tr><td></td><td>variable (measured)</td><td>attributive (gauge)</td><td></td></tr><tr><td></td><td>attributive (gauge)</td><td>attributive (gauge)</td><td></td></tr><tr><td></td><td>variable (measured)</td><td>variable (measured)</td><td></td></tr><tr><td></td><td>100% (in prod. line)</td><td>100% (in production line)</td><td></td></tr><tr><td></td><td>Ppk ≥ 1,67</td><td>variable (measured)</td><td></td></tr><tr><td></td><td>Ppk ≥ 1,67</td><td>Cpk ≥ 1,33</td><td></td></tr><tr><td>special character</td><td>for PPAP only</td><td>for series production</td><td></td></tr></table>			Significant characteristics		100% (in prod. line)	100% (in prod. line)			Ppk ≥ 2,00	Cpk ≥ 1,67			Ppk ≥ 1,67	attributive (gauge)			variable (measured)	attributive (gauge)			attributive (gauge)	attributive (gauge)			variable (measured)	variable (measured)			100% (in prod. line)	100% (in production line)			Ppk ≥ 1,67	variable (measured)			Ppk ≥ 1,67	Cpk ≥ 1,33		special character	for PPAP only	for series production	
		Significant characteristics		100% (in prod. line)		100% (in prod. line)																																							
				Ppk ≥ 2,00		Cpk ≥ 1,67																																							
	Ppk ≥ 1,67		attributive (gauge)																																										
	variable (measured)		attributive (gauge)																																										
	attributive (gauge)		attributive (gauge)																																										
	variable (measured)		variable (measured)																																										
	100% (in prod. line)		100% (in production line)																																										
	Ppk ≥ 1,67		variable (measured)																																										
	Ppk ≥ 1,67		Cpk ≥ 1,33																																										
special character	for PPAP only		for series production																																										
Characteristics, which are required below (critical or significant characteristic) have to be also implemented in all other Q-documents like FMEA, Control Plan, etc.																																													
	When to apply a C? Critical characteristic Product features or process features, which affect the compliance with safe operation or Government regulations	General	Have to be marked continuously in the FMEA's and controlplan. (Basically Melecs specified symbol has to be used unless Melecs defined in the technical documents other symbols). Labeling of C-characteristic - see SC table above. C's have to be evaluated in FMEA acc. newest handbook rules. Capability and measurement system analyses MSAs of associated measuring equipment have to be enclosed to the initial sampling report to Melecs Deviation approval possible in case out of spec.? No When suppliers have their own special characteristic classification, then a translation table (supplier classification vs. Melecs/customer classification) has to be available																																										
		Initial sampling*	Preliminary process capability / machine capability pp, ppk ≥ 2,0 / cm, cmk ≥ 2,0, min. 50 samples in a row (+- 6s, 0,002 ppm) (per cavity) Is the process not capable then it must be a 100% check until the process has been optimised. Measurement system analyses/measurement process analyses has to be performed according VDA Vol. 2 or MSA last revision, arranged with responsible SQA. cg, cgk ≥ 1,33 / %R&R <10% (procedure depends on the measuring process) Confirmation by CoC (Certificate of Conformity) document with each delivery (after green PPAP).																																										
		Serial production*	<u>Testing methode SPC:</u> In production a SPC control of characteristic has to be performed long-term process capability cp, cpk ≥ 1,67, min. 125 pcs., 25 spot check to 5 pcs. (+- 5s, 0,6 ppm) (per cavity) Confirmation by CoC (Certificate of Conformity) document with each delivery. <u>Poka-Yoke</u> <u>Testing methode 100%:</u> 100% automatical check of characteristic Generally to all testing methodes records have to be made and if necessary send to MELECS, also applies for material certificates.																																										

* the capability requirements are the general requirements, if there are no special requirements from Melecs than the given values are valid